

A novel rate splitting scheme for uplink cognitive radio networks

P.T.Q. Trang^{1,2}, H.X. Son¹, B.N. Son¹, D.T. Duong¹, T.A. Vu¹

Abstract—Cognitive radio (CR) combined with rate-splitting multiple access (RSMA) provides a promising paradigm for enhancing spectrum utilization while guaranteeing primary user (PU) protection. In this letter, we present a CR-RSMA framework that addresses three critical aspects of spectrum sharing. First, we establish the feasibility condition under which a secondary user (SU) can be paired with a PU to share the same resource block. Second, we introduce a rate-splitting based decoding technique at the base station to effectively separate PU–SU signals once the pairing condition is satisfied. Third, we develop an algorithm that maximizes the number of feasible PU–SU pairs, thereby demonstrating the efficiency of the proposed scheme. Numerical results confirm that the proposed method improves the number of feasible PU–SU pairs up to 50% when compared to the CR-NOMA while ensuring QoS guarantees for PUs.

Index Terms—cognitive radio, RSMA, NOMA, SIC, UAV.

I. INTRODUCTION

THE sixth-generation (6G) wireless networks are envisioned to support massive connectivity for heterogeneous devices such as unmanned aerial vehicles (UAVs), sensors, wearables, and IoT terminals, many of which operate in unlicensed bands. Such large-scale access intensifies spectrum scarcity and interference to licensed users, motivating cognitive radio (CR) as an enabler of dynamic spectrum sharing [1]. In parallel, integrated sensing and communication (ISAC) has been identified as a core 6G pillar, jointly exploiting spectrum for communication and sensing, which further highlights the need for interference-aware and flexible access schemes [2]. While CR enables unlicensed devices to access licensed spectrum dynamically, its performance in dense deployments remains constrained by inter-user interference and static resource allocation strategies. Non-orthogonal multiple access (NOMA) has been recognized as a multiple access scheme to further improve spectral efficiency by enabling multiple users to share the same frequency resources with different power allocation levels [3]. The integration of CR and NOMA leverages the dynamic spectrum access capability of CR and the high spectral efficiency of NOMA, thereby offering enhanced connectivity and throughput for heterogeneous wireless networks [4].

Research on cognitive radio-inspired NOMA has established explicit pairing conditions under which a primary user (PU, a licensed user) and a secondary user (SU, an unlicensed user) can share the same resource block, ensuring both spectrum reuse and PU protection via carefully designed interference

thresholds and power control [5]. Moreover, hybrid successive interference cancellation (SIC) schemes enable the base station (BS) to reliably strip off the SU’s signal while decoding the PU, achieving zero-error probability (i.e., floor error probability of zero) for the SU when SIC succeeds [6]. Nevertheless, traditional CR-NOMA does not fully exploit the entire capacity region (i.e., the full achievable DoF region), leaving performance on the capacity boundary unutilized [7]. To address this limitation, recent studies propose the integration of rate-splitting multiple access (RSMA) into CR-NOMA systems. In such CR-RSMA frameworks, the SU splits its message, allowing part of the interference to be decoded and part to be treated as noise, thereby fully utilizing the available capacity region and substantially improving SU outage performance without harming PU’s QoS [7]–[9].

However, existing studies on CR-RSMA have primarily focused on performance comparison against CR-NOMA or on evaluating outage probabilities of RSMA systems, without providing conditions for PU–SU pairing or analytical guarantees for PU protection. Moreover, the optimal rate-splitting, which critically determines both PU reliability and SU throughput, remains largely unexplored in a tractable analytical form. This gap hinders the systematic design of CR-RSMA in practical heterogeneous 6G deployments [10]. In this letter, we present a CR-RSMA framework that can address three critical aspects of spectrum sharing. First, we establish the feasibility condition under which a SU can be paired with a PU to share the same resource block. Second, we introduce a rate-splitting based decoding technique at the base station (BS) to effectively separate PU–SU signals once the pairing condition is satisfied. Third, we develop a resource-aware algorithm that maximizes the number of feasible PU–SU pairs, thereby demonstrating the efficiency of the proposed scheme.

II. SYSTEM MODEL AND PROBLEM STATEMENT

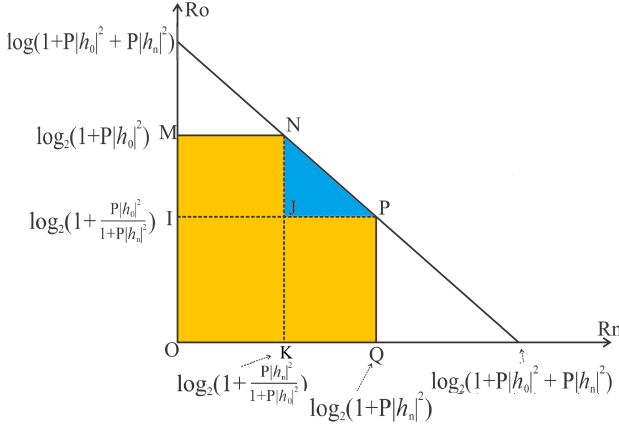
We first consider an uplink system consisting of a primary user, denoted by U_0 , and a secondary user, denoted by U_n . In practice, the PU may represent a ground-based mobile device, while the SU can be a UAV acting as an aerial user. UAV-based SUs are attractive due to their flexible deployment, extended line-of-sight (LoS) links, and enhanced coverage in areas with limited terrestrial communications.

The transmit powers of the PU and SU are fixed at P_0 and P_n , respectively. The complex channel coefficients from U_0 and U_n to the base station (BS), estimated through pilots, are denoted by h_0 and h_n . In addition, the PU’s rate requirement R_0 must be guaranteed, while the SU’s achievable rate must exceed a given threshold R_s .

Based on this model, the cognitive radio (CR) problems are stated as follows:

¹D.T. Duong, T.A. Vu are lecturers and P.T.Q. Trang, H.X. Son, B.N. Son are Ph.D. students with the Department of Electrical Engineering and Communication, VNU University of Engineering and Technology, Ha Noi, Viet Nam. Sonxuanha@yahoo.com.vn, sonttkxtb@gmail.com, duongdt@vnu.edu.vn, vuta@vnu.edu.vn

²P.T.Q. Trang is with the Faculty of Electrical Engineering and Communication, Ha Noi University of Industry, Viet Nam. pham.trang@hau.edu.vn

Fig. 1. Capacity region graph for each pair (R_S, R_0) .

- 1) Given R_0, R_s, P_0, h_0, P_n , what condition must the SU's channel h_n satisfy so that the SU can be admitted to share the same spectrum with the PU? We refer to this as the *PU-SU pairing condition*.
- 2) Once the SU satisfies the pairing condition, **how can the BS separate the messages from the PU and SU?**
- 3) When extending to scenarios with multiple randomly distributed PUs and SUs, how does the pairing efficiency vary under different pairing conditions?

III. PROPOSED APPROACH

This section presents a CR-RSMA framework for efficient PU-SU spectrum sharing. A closed-form feasibility condition is derived, followed by a signal separation strategy and an admission-based algorithm to maximize admitted SU-PU pairs under resource constraints.

A. Conditions for SU-PU Resource Sharing

Based on the achievable rate region of the two-user uplink multiple access channel as illustrated in Fig. 1 [11], the SU can be paired with the PU if and only if the target rate point $T(R_s, R_0)$ lies within the feasible pentagon region $OMNPQ$, which is bounded by the Shannon capacity. Without loss of generality, we assume equal transmit power for both users, i.e., $P_0 = P_n = P$. Therefore, the following condition must be satisfied:

$$R_s + R_0 < R_n + R_0 = \log_2(1 + P|h_0|^2 + P|h_n|^2). \quad (1)$$

From the sum-rate constraint, we obtain

$$P|h_n|^2 > 2^{R_s + R_0} - P_0|h_0|^2 - 1. \quad (2)$$

On the other hand, the individual rate constraint of the SU requires

$$R_s < \log_2(1 + P|h_n|^2) \Leftrightarrow P|h_n|^2 > 2^{R_s} - 1. \quad (3)$$

By combining these two conditions, the proposed criterion for admitting an SU to share the spectrum with a PU can be expressed as

$$P|h_n|^2 > \max(2^{R_s} - 1, 2^{R_s + R_0} - P_0|h_0|^2 - 1). \quad (4)$$

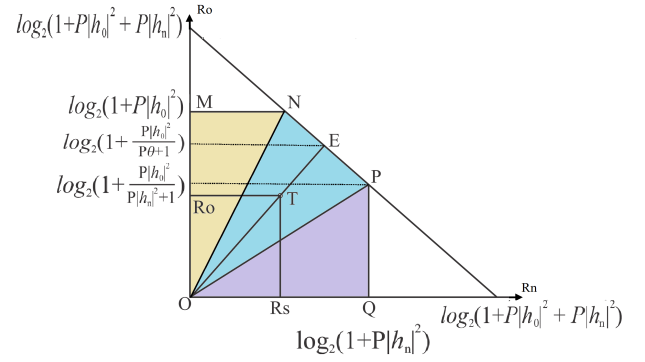


Fig. 2. The capacity graph .

It is worth noting that, to the best of our knowledge, such a concise pairing condition has not been explicitly reported in prior works.

B. Rate-Splitting and SIC Feasibility

Determining the appropriate rate-splitting plays a crucial role in RSMA, as it directly impacts the achievable rates of users, enhances spectral efficiency, and ensures reliable decoding performance at the receiver [12]. This subsection introduces the proposed PU-SU separation and suitable rate-splitting in the CR-RSMA scheme.

The capacity region of the PU-SU pair, illustrated in Fig. 2, is partitioned into three triangular subregions: the yellow area ($\triangle OMN$), the blue area ($\triangle NOP$), and the purple area ($\triangle POQ$). Each subregion corresponds to a particular decoding condition and represents a distinct relationship between the achievable rates of the PU and the SU.

Let $T(R_0, R_s)$ denote the point corresponding to the required rate pair with the channel gains $|h_0|^2$ and $|h_n|^2$ of the PU and SU, which are assumed to be perfectly known through pilot transmission broadcast. Based on this information, BS identifies a specific triangular region in which point T is located. The determination of the location of point T is essential for selecting the appropriate decoding strategy for the (PU, SU) pair. To classify the region where T is located, only compare the following three parameters: $\tan(\angle TOQ) =$

$$\frac{R_0}{R_s}, \quad \tan(\angle NOQ) = \frac{\log_2\left(1 + \frac{P|h_0|^2}{1}\right)}{\log_2\left(1 + \frac{P|h_n|^2}{P|h_0|^2 + 1}\right)}, \quad \tan(\angle POQ) = \frac{\log_2\left(1 + \frac{P|h_0|^2}{P|h_n|^2 + 1}\right)}{\log_2\left(1 + \frac{P|h_n|^2}{1}\right)}.$$

In the first case, where $\tan(\angle TOQ) > \tan(\angle NOQ)$, the point T located inside the triangular region OMN . Hence, T belongs to the NOMA region. BS proceeds to separate the N -rate point according to the PU-preferential NOMA technique as follows:

$$R'_0 = \log_2\left(1 + \frac{P|h_0|^2}{P|h_n|^2 + 1}\right), \quad R_n = \log_2\left(1 + \frac{P|h_n|^2}{1}\right) \quad (5)$$

It can be observed that $R'_0 > R_0$ and $R_n > R_s$, thereby ensuring this separation technique is feasibility.

In the second case, when $\tan(\angle TOQ) < \tan(\angle POQ)$, the point T falls within the triangular region POQ , which also belongs to the NOMA region. The BS separates point P via CSI-based NOMA:

$$R_n = \log_2 \left(1 + \frac{P|h_n|^2}{P|h_o|^2 + 1} \right), \quad R'_o = \log_2 \left(1 + \frac{P|h_o|^2}{1} \right) \quad (6)$$

It can also be observed that $R'_0 > R_0$ and $R_n > R_s$, thereby ensuring this separation technique is feasibility.

From the above analysis, it can be observed that for both triangular subregions OMN and POQ (the yellow and purple areas in Fig.2) located within the conventional NOMA domain, the decoding order is straightforwardly ensured by NOMA. In these cases, no rate-splitting needs to be determined, as both PU and SU can be successfully decoded without message splitting.

In contrast, for the point T lies inside the triangular region NOP . Unlike the previous two cases, this region cannot be entirely supported by the conventional NOMA decoding order. Therefore, a rate-splitting strategy is required to guarantee successful decoding for both PU and SU. Specifically, to ensure reliable decoding of the (PU, SU) pair, we divided the SU's message into a common part and a private part, such that the common stream is decoded first, followed by the private messages of PU and SU. Hence, determining the rate-splitting is essential within this RSMA subregion.

Consider the case where point T lies inside the triangle ONP . Extending line OT intersects NP at point E with coordinates (x, y) . We use the coordinates of E to determine the rate-splitting ratio for the SU. Specifically, $\tan \angle EOQ = R_0/R_s = y/x$. Since E lies on line NP , its coordinates (x, y) satisfy the equation of NP . Hence, (x, y) can be obtained as the solution of the following system:

$$\begin{cases} y = \frac{R_0}{R_s} x \\ y = -x + \log_2(1 + P|h_0|^2 + P|h_n|^2) \end{cases} \quad (7)$$

By solving the system of equations in (7), we obtain:

$$y = \frac{\log_2(1 + P|h_0|^2 + P|h_n|^2)}{1 + \frac{R_s}{R_0}}. \quad (8)$$

For analytical tractability, we further define:

$$y = \log_2 \left(1 + \frac{P|h_0|^2}{P\theta + 1} \right). \quad (9)$$

where θ denotes the rate-splitting in our proposed CR-RSMA model. From (9), we obtain a closed-form expression for the rate-splitting as:

$$\theta = \frac{|h_0|^2}{2^y - 1} - \frac{1}{P}. \quad (10)$$

The practicality of the threshold in (10) is validated by showing that the corresponding operating point enables feasible SIC at the base station.

Since point E is on the line NP , its ordinate is bounded between those of points P and N , which implies:

$$P|h_n|^2 > P\theta \quad (11)$$

The received SU component $P|h_n|^2$ can be partitioned into two virtual parts: $P|h_n|^2 - P\theta$ and $P\theta$. Consequently, the base station observes three effective signal components. To recover the intended messages, the successive interference cancellation technique is applied in the following decoding order: $P|h_n|^2 - P\theta$, $P|h_0|^2$ and $P\theta$. The corresponding achievable rates are given by:

$$R_1 = \log_2 \left(1 + \frac{P|h_n|^2 - P\theta}{P\theta + P|h_0|^2 + 1} \right);$$

$$R_2 = \log_2 \left(1 + \frac{P|h_0|^2}{P\theta + 1} \right); R_3 = \log_2 \left(1 + \frac{P\theta}{1} \right). \quad (12)$$

For successful SIC decoding, it must hold that the achievable rate of the primary signal is no less than the target rate R_0 , and the combined achievable rates allocated to the secondary signal exceed R_s . From expressions (11) and (12), it can be verified that the conditions $R_0 < R_2$ and $R_s < R_1 + R_3$ are always satisfied. Hence, the selected rate-splitting ensures the feasibility of SIC decoding in the considered CR-RSMA framework. It is worth noting that the proposed technique explicitly characterizes the partitioning of the region $P\theta$, which, to the best of our knowledge, has not been explicitly addressed in prior RSMA publications.

Outage Probability Analysis:

Let $\mathbb{P}_1, \mathbb{P}_2$, and $\mathbb{P}_3$ denote the outage probabilities corresponding to the case where the operating point T lies within the feasible regions represented by triangles OMN , NOP , and POQ , respectively. Accordingly, the overall outage probability of user Un can be expressed as:

$$\begin{aligned} \mathbb{P}_{outage} = & \mathbb{P}_1(\log_2 \left(1 + \frac{P|h_n|^2}{P|h_0|^2 + 1} \right) < R_s) + \\ & \mathbb{P}_2(\log_2 \left(1 + P|h_n|^2 \right) < R_s) + \\ & \mathbb{P}_3(\log_2 \left(1 + \frac{P|h_0|^2 - P\theta}{P\theta + P|h_0|^2 + 1} \right) + \log_2 \left(1 + \frac{P\theta}{1} \right) < R_s). \end{aligned} \quad (13)$$

It can be seen that $\mathbb{P}_2 = \mathbb{P}_3 = 0$ when $P \rightarrow \infty$.

Consider probability $\mathbb{P}_1$. Since the point T lies within the triangle ONP , so:

$$\frac{R_0}{R_s} > \frac{\log_2(1 + P|h_0|)}{\log_2 \left(1 + \frac{P|h_n|}{1 + P|h_0|} \right)}. \quad (14)$$

From inequality (14) leads to:

$$\begin{aligned}
 & \mathbb{P}_1 \left(\log_2 \left(1 + \frac{P|h_n|^2}{P|h_0|^2 + 1} \right) < R_s \right) \\
 &= \mathbb{P}_1 \left(\frac{1}{\log_2 \left(1 + \frac{P|h_n|^2}{P|h_0|^2 + 1} \right)} > \frac{1}{R_s} \right) \\
 &= \mathbb{P} \left(\frac{\log_2 (1 + P|h_0|^2)}{\log_2 \left(1 + \frac{P|h_n|^2}{P|h_0|^2 + 1} \right)} > \frac{\log_2 (1 + P|h_0|^2)}{R_s} \right) \\
 &< \mathbb{P} \left(\frac{R_0}{R_s} > \frac{\log_2 (1 + P|h_0|^2)}{R_s} \right) = \\
 &\mathbb{P} \left(R_0 > \log_2 (1 + P|h_0|^2) \right) = \mathbb{P}(|h_0|^2 < (2^{R_0} - 1)/P). \\
 &\text{This probability approaches zero as } P \rightarrow \infty. \quad (15)
 \end{aligned}$$

C. Admission Control Algorithm

In this section, the CR network scenario is extended as illustrated in Fig. 3. A cognitive radio system with N mobile phones as primary users and M UAVs as secondary users is considered. The channel gains of both PUs and SUs are assumed to be randomly distributed within the feasible regions defined by: $2^{R_0} - 1 < P|h_0|^2 < 2^{R_0+R_s} - 1$ for PU links and $2^{R_s} - 1 < P|h_m|^2 < 2^{R_0+R_s} - 1$ for SU links. Based on these random realizations, an efficient pairing and admission strategy is required to **maximize the number of admitted SUs while guaranteeing the QoS of the primary system**. To this end, the proposed algorithm exploits the pairing condition established in the previous subsection and employs a sorting-based mechanism to match SUs with PUs under the given constraints.

The procedure to determine the number of candidates that can be admitted to share resources with U_0 is summarized in Algorithm 1. In this algorithm, the PU channel gains are first sorted in descending order, while the SU channel gains are sorted in ascending order. For each SU, a for-loop iteratively checks the available PUs from the highest to the lowest channel gain to identify a feasible pair; once a pair is found, the loop terminates for that SU. Although this approach does not guarantee global optimality, it provides a fair and consistent basis for comparing the performance of different pairing strategies under the same framework.

IV. NUMERICAL RESULT AND DISCUSSION

In this section, we present simulation results to validate the proposed approach. First, we evaluate the admission condition in (4) for PU-SU pairing. Then, demonstrate the feasibility of the decoding order in CR-RSMA and provide comparisons with CR-NOMA. These results provide insights into both the feasibility and the performance advantages of the proposed scheme.

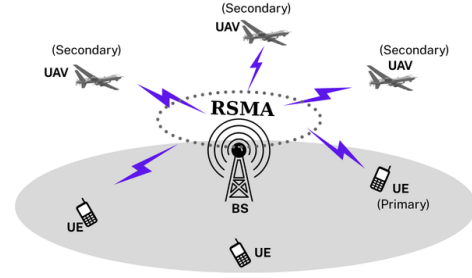


Fig. 3. Expanded scenario with multiple randomly distributed PUs and SUs.

Algorithm 1 Algorithm to maximize the paired PU, SU

Input: Number of users N , primary rate R_0 , set of secondary rates R_s
Output: Number of successful couples (PU, SU) $\leftarrow C$

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for  $r = 1$  to  $\text{length}(R_s)$  do
   $A \leftarrow 2^{R_0+R_s[r]} - 1$ 
  for  $i = 1$  to  $N$  do
     $Po[i] \leftarrow (2^{R_0} - 1) + (A - (2^{R_0} - 1)) \cdot \text{rand}(0, 1)$ 
     $Pm[i] \leftarrow (2^{R_s[r]} - 1) + (A - (2^{R_s[r]} - 1)) \cdot \text{rand}(0, 1)$ 
  end for
  Sort  $Po$  in ascending order
  Sort  $Pm$  in descending order
  Initialize counter  $u \leftarrow 0$ 
  for  $i = 1$  to  $N$  do
    for  $k = 1$  to  $N$  do
      if  $Pm[k] > \max(2^{R_s[r]} - 1, A - Po[i])$  then
         $u \leftarrow u + 1$ 
         $Pm[k] \leftarrow 0$ 
        break
      end if
    end for
    end for
   $C[r] \leftarrow u$ 
end for

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In the first scenario, Algorithm 1 (Subsection III-c) is used for PU-SU pairing. System parameters are summarized in Table I. The channel is modeled based on QoS requirements, with P representing the required SINR for each user, randomly drawn between the minimum SINR for R_0 and the total rate $R_0 + R_s$. This scenario validates condition (11), the admission criterion for an SU to share the spectrum with a PU.

Fig.4 illustrates the number of admitted PU-SU pairs versus the secondary rate requirement R_s under different system configurations. When the number of SUs equals the number of PUs ($N_{PU} = N_{SU} = 10$) and the $R_0 = 0.5$, the proposed scheme consistently outperforms the CR-NOMA benchmark, maintaining nearly full admission capability even as R_s increases. In contrast, CR-NOMA rapidly loses feasible pairs as the SU rate grows, highlighting the advantage of the proposed rate-splitting strategy in preserving user admission.

When the number of SUs exceeds the number of PUs ($N_{PU} = 10, N_{SU} = 20$), both schemes experience a performance degradation due to the increased competition among SUs. Nevertheless, the proposed method still sustains a higher

TABLE I
THE SIMULATION PARAMETERS

Parameters	Value
Number of PUs (N)	10
Number of SUs (N)	(10, 20)
Rate of primary (R_0)	(0.5, 1)
Rate of secondary (R_S)	0.5:5

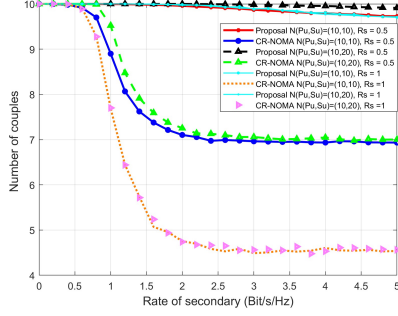


Fig. 4. Number of PU-SU pairs paired to share resources.

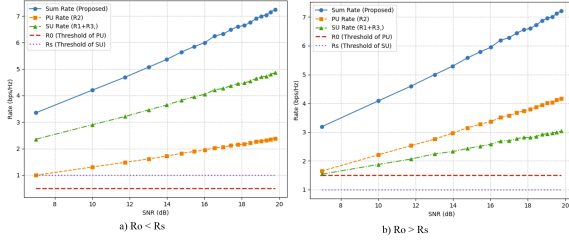


Fig. 5. The rate of PU and SU.

number of admitted pairs compared to CR-NOMA, demonstrating better scalability with user density.

Finally, when $R_0 = 1$, the performance gap becomes even more pronounced. Specifically, while CR-NOMA experiences a sharp degradation in the number of feasible pairs as the secondary rate increases (dropping to nearly 5 pairs when $R_s = 1$), the proposed method is able to sustain close to the maximum number of pairs across the entire range. In particular, the proposed scheme achieves up to a 50% increase in the pairing ratio compared with CR-NOMA when $R_s = 1$. This confirms that the proposed admission and rate-splitting framework is more robust to stringent QoS requirements than CR-NOMA.

Fig.5 shows the achievable rates of the PU and SU under the proposed CR-RSMA scheme, where the SU rate-splitting ratio and the SIC decoding order are determined by our design. Two scenarios are considered: (a) SU prioritized ($R_0 < R_S$) and (b) PU prioritized ($R_0 > R_S$). In both cases, the QoS requirements are satisfied, as the SU rate remains above R_S and the PU rate exceeds R_0 . Moreover, in scenario (a), where the SU is prioritized, the SU achieves a higher rate than the PU, while in scenario (b), prioritizing the PU leads to consistently higher PU rates compared to the SU. These results confirm the flexibility of the proposed scheme in adapting to different priority settings by properly selecting θ and the SIC order.

V. CONCLUSION

This letter introduced a CR-RSMA framework for uplink cognitive radio networks, providing a closed-form admission condition, explicit rate-splitting threshold, and an admission-based pairing algorithm. Simulation results confirmed significant gains over CR-NOMA in terms of pairing efficiency. Furthermore, by jointly optimizing the rate-splitting and SIC order, the proposed framework ensures PU QoS while flexibly adapting to different priority settings. Beyond terrestrial networks, the proposed framework can be extended for efficient spectrum sharing in UAV-assisted communications, ISAC systems, and future wireless networks.

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